CHARMM-GUI

Effective Simulation Input Generator and More

CHARMM is a versatile program for atomic-level simulation of many-particle systems, particularly macromolecules of biological interest. - M. Karplus

Some lectures, job postings, and FAQ are now available. See update log for update history and giving for donation. Contact info is given below.

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Bilayer Builder

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PDB Info | STEP 1 | STEP 2 | STEP 3 | STEP 4 | STEP 5 | STEP 6

JOB ID: 3757172590

Original PDB File:	8HBV_WITH_8G8W_CARDIOLIPINS.pdb (view structure)	download.tgz
Individual Chains:	8hbv_with_8g8w_cardiolipins_proa.pdb 8hbv_with_8g8w_cardiolipins_heta.pdb	
CHARMM Input:	step1_pdbreader.inp	
CHARMM Output:	step1_pdbreader.out	
CHARMM PDB:	step1_pdbreader.pdb (view structure)	
CHARMM CRD:	step1_pdbreader.crd	
CHARMM PSF:	step1_pdbreader.psf	

Computed Energy:

Please beware of that the computed energy is CHARMM single-point energy and is displayed to make sure all the coordinates are defined.

```
ENER ENR:  Eval#     ENERgy      Delta-E         GRMS
ENER INTERN:          BONDs       ANGLes       UREY-b    DIHEdrals    IMPRopers
ENER CROSS:           CMAPs        PMF1D        PMF2D        PRIMO
ENER EXTERN:        VDWaals         ELEC       HBONds          ASP         USER
 ----------       ---------    ---------    ---------    ---------    ---------
ENER>        0 5149906.4898       0.0000  506389.0980
ENER INTERN>      597.38172    799.71184    138.94908   2881.13454      3.01852
ENER CROSS>        -8.22824      0.00000      0.00000      0.00000
ENER EXTERN>   5148467.9352   -2973.4129       0.0000       0.0000       0.0000
 ----------       ---------    ---------    ---------    ---------    ---------
```

Orientation Options: ?

○ Use PDB Orientation	This option is suggested for an oriented structure from http://opm.phar.umich.edu
○ Align the First Principal Axis Along Z	This option is suggested for small helical bundle or homo-oligomer.
○ Align a Vector (Two Atoms) Along Z	This option is suggested for an irregular, hetero-oligomer.
◉ Run PPM 2.0	This option run executable for given input structure at https://opm.phar.umich.edu/ppm_server2_cgopm It may take some minutes depending on protein size.

Please select the chains sending to PPM server.

☑ PROA

Positioning Options:

☐ Rotate Molecule respect to the X axis [ ] Degree

☐ Rotate Molecule respect to the Y axis [ ] Degree

☐ Translate Molecule along Z axis [ ] Angstrom

☐ Flip Molecule along the Z axis

Area Calculation Options:

☑ Generate Pore Water and Measure Pore Size

- ◉ Using protein geometry Water molecules outside the XY extent of the transmembrane domain would be removed.
- ○ Using a cylindrical radius of ______ Angstrom

Next Step:
Calculate Cross-Sectional Area